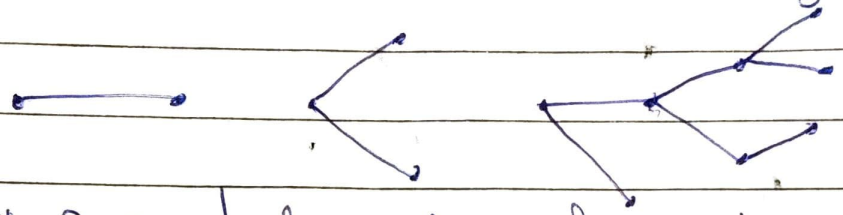
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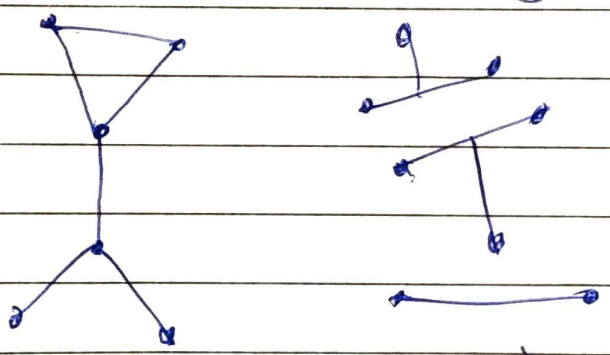
Unit-5

Trees :-

A graph G is said to be a tree if it is connected and has no cycles.



* A pendant vertex of a tree is also called leaf.



(a) (b)

Such type of graphs are called forest. The graph which is tree is usually denoted by T.

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Theorem 1: In a tree, there is one and only one path between every pair of vertices.

Proof: Let T be a tree, Then T is a connected simple graph. Since T is connected, there must be at least one path between every two vertices. If there are two paths between a pair of vertices of T, the union of the paths will become a cycle, and T cannot be a tree. Thus, between every pair of vertices in a tree there must exist one and only one path.

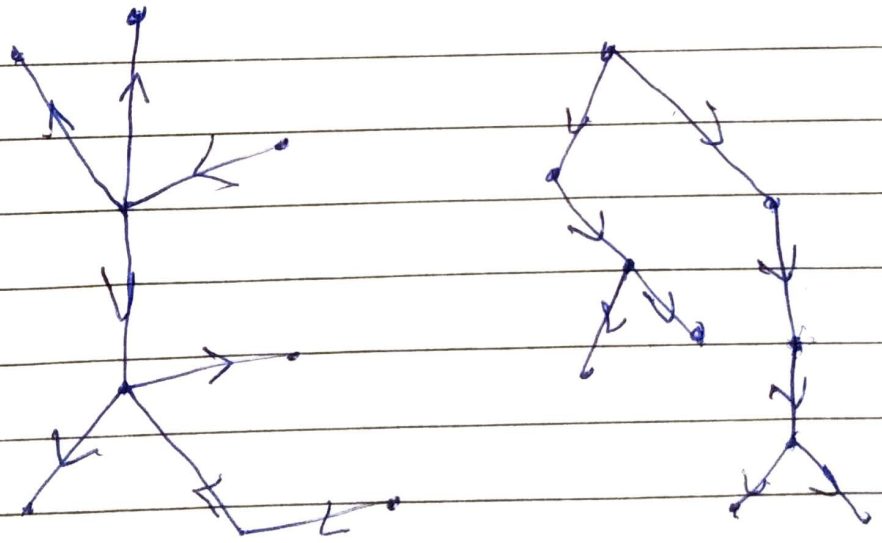
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Rooted tree :-

Let D be a directed graph and G be its underlying graph. We say that D is a directed tree whenever G is a tree. A directed tree T is called a rooted tree.

if (i) T contains a unique vertex, called the root whose in-degree is Equal to 0,

(ii) The in-degrees of all other vertices of T are Equal to 1.



m-ary-tree

A rooted tree T is called an m-array tree if every vertex of T is of out-degree $\leq m$ that is if every of vertex of T has at most m children.

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A rooted tree T is called a complete m-ary tree if every internal vertex of T is of out-degree m, that is if every internal vertex of T has exactly m children.

Binary Tree :-

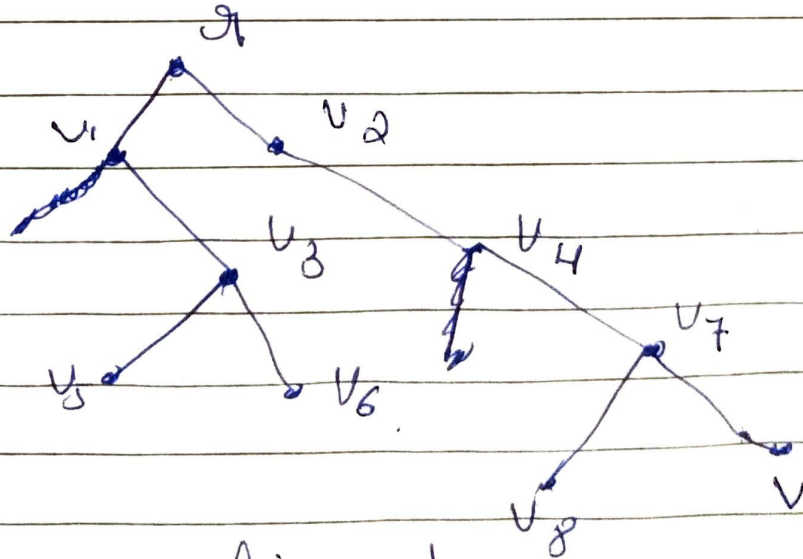
An m-ary tree for which m = 2 is called a binary tree.

or

A rooted tree T is called a binary tree if every vertex of T is of out-degree ≤ 2, that is if every vertex has at most two children.

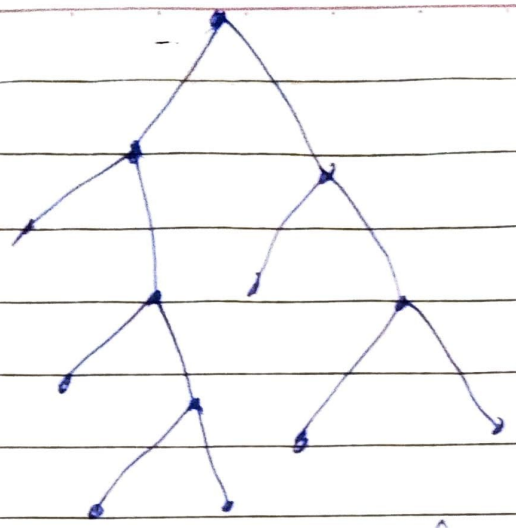
A complete m-ary tree for which m = 2 is called a complete binary tree.

Ex :-



Binary tree.

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Complete binary tree

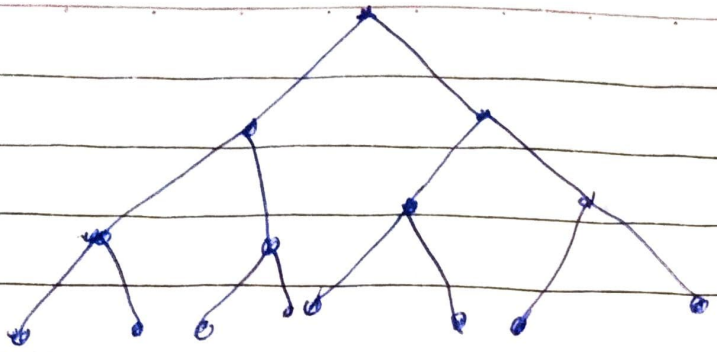
Balanced Tree :-

If T is a rooted tree and h is the largest leaf number achieved by a leaf of T, then T is said to have height h. A rooted tree of height h is said to be balanced if the leaf number of every leaf is h or h-1.

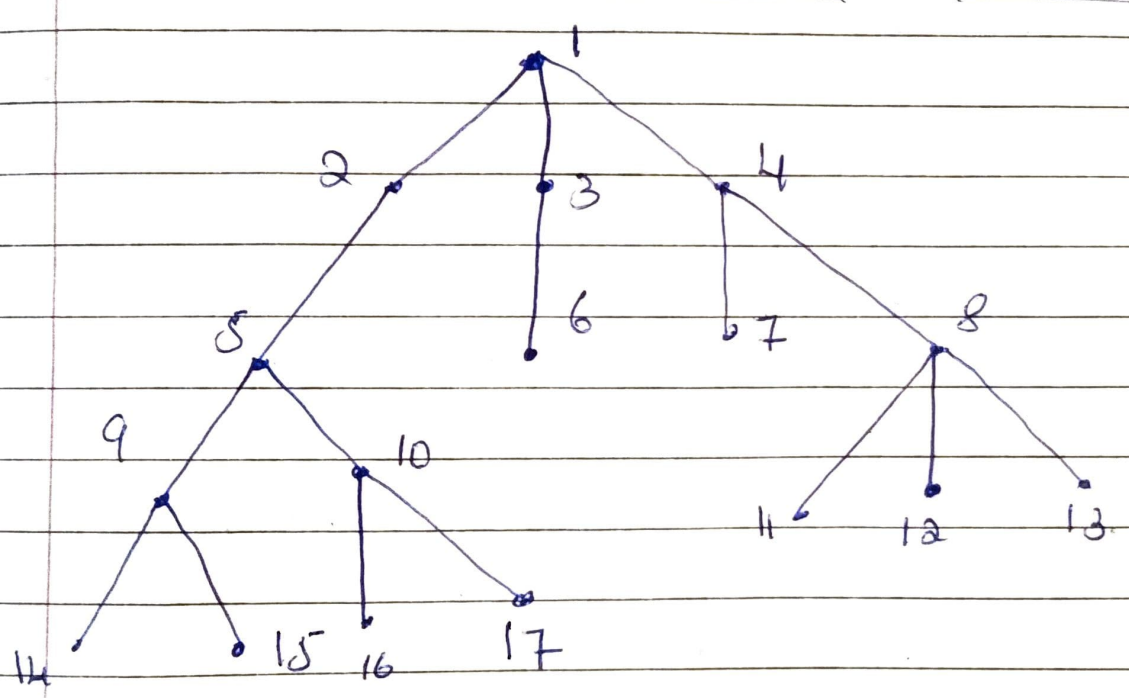
Full binary Tree

Let T be a Complete binary tree of height h, Then T is called a full binary tree if all the leaves in T at level h

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Preorder and Postorder traversals.



Preorder :- 1, 2, 5, 9, 14, 15, 10, 16, 17 3, 6, 4, 7, 8, 11, 12, 13.

Suppose we list the vertices of T such that r is the first vertex and the subsequent vertices are selected in the following order.

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Vertices of T_1 read from left to right, Vertices of T_2 read from left to right. The list of vertices so Prepared is called the Preorder traversal of T

Suppose we list the vertices of T scanned in the following order. Vertices of T_1 read from the leftmost descendant, Vertices of T_2 read from the leftmost descendant Vertices of T_3 read from the leftmost-descendant, and so on and finally the root. The list of vertices so Prepared is Called the Postorder Traversal of T.

Postorder :- 14, 15, 9, 16, 17, 10, 5, 2, 6, 3, 7, 11, 12, 13, 8, 4, 1

Sorting :- It consist of two Parts.

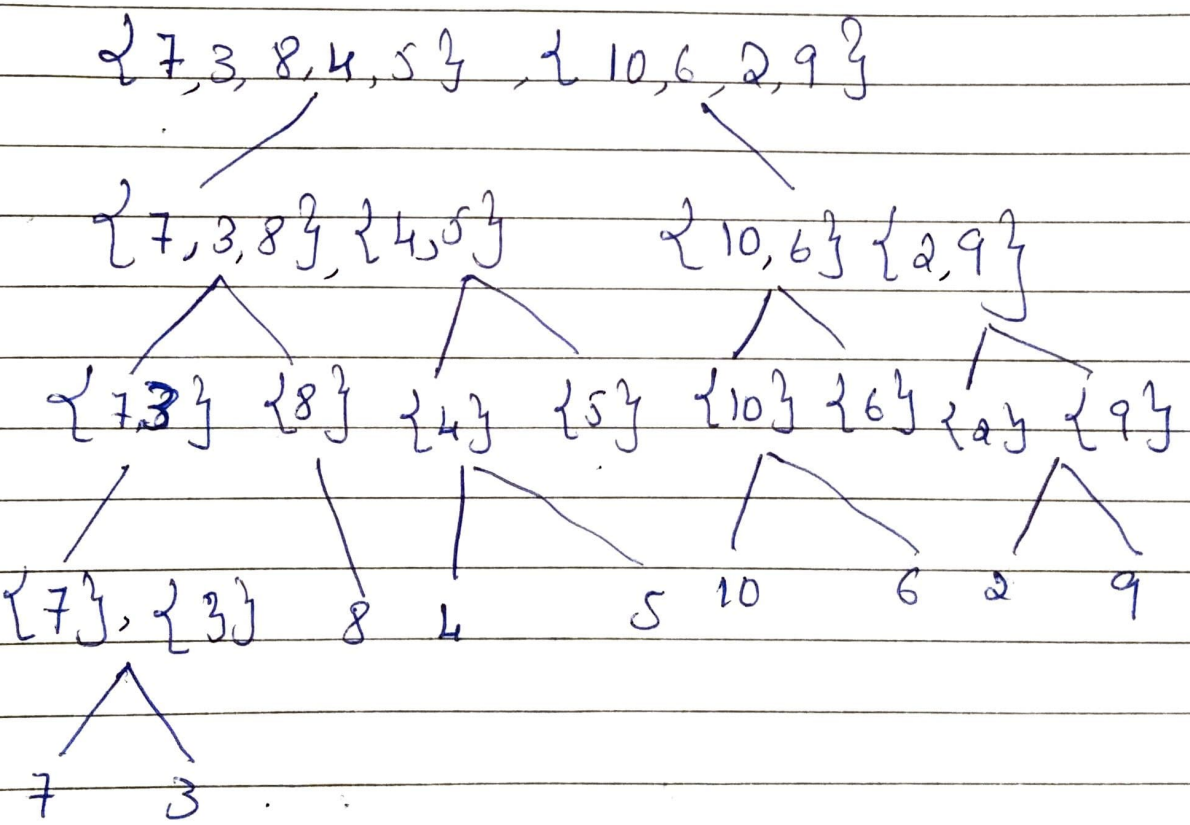
(1) Split the given list and all subsequent lists in half until Each Sublist Contains a single element.

(2) Merge the sublists in nondecreasing order until the original n integers have

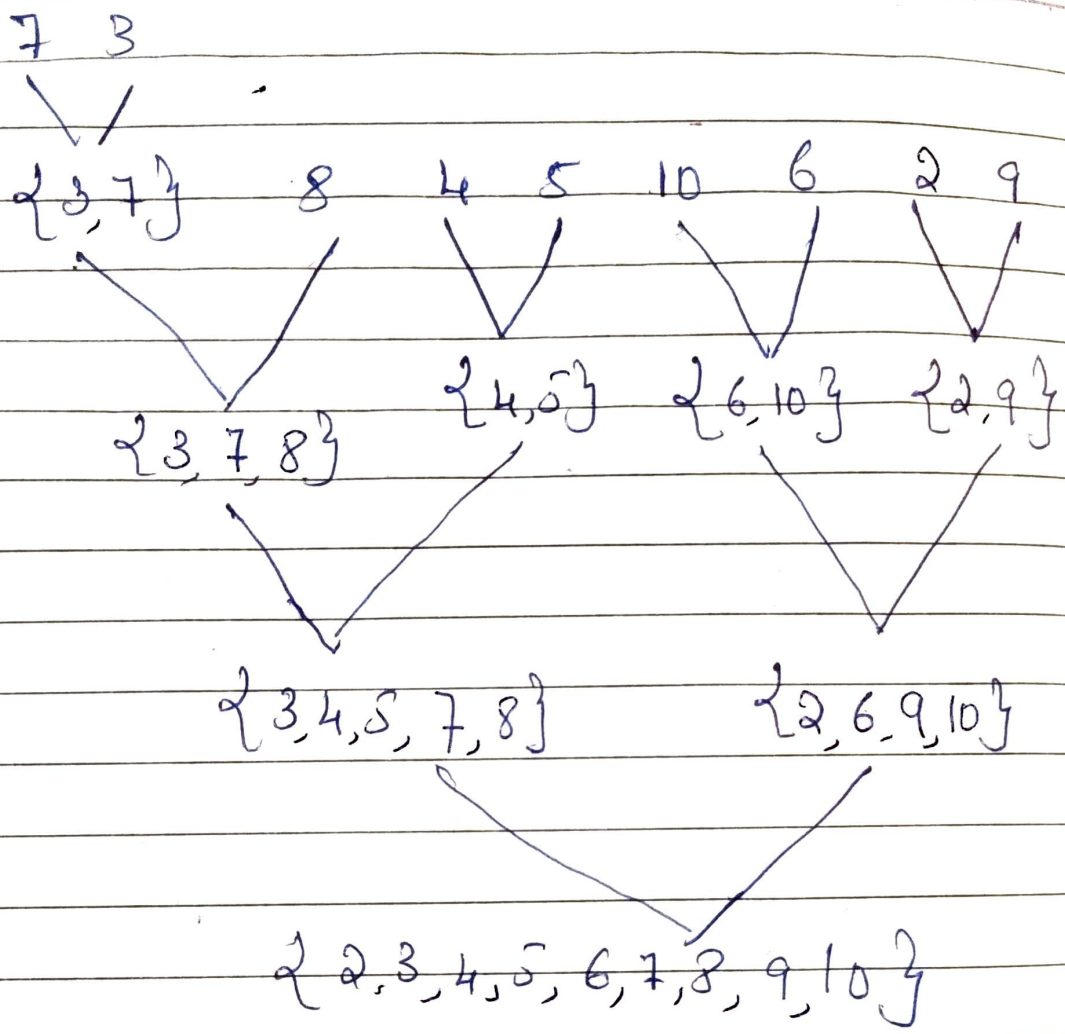
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Then sorted. Splitting and merging process is done by the use of balanced complete trees. This method of sorting a list is known as merge sort.

(1) Using the merge-sort method, sort the list 7, 3, 8, 4, 5, 10, 6, 2, 9.

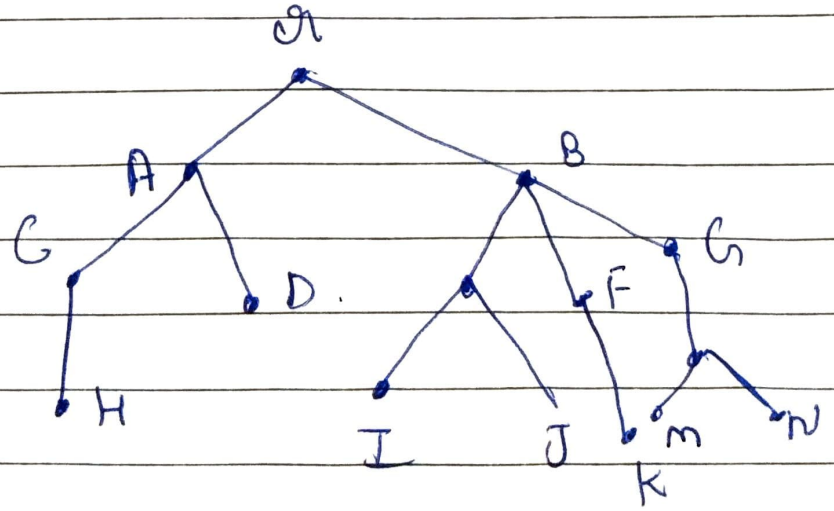


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Problems :-

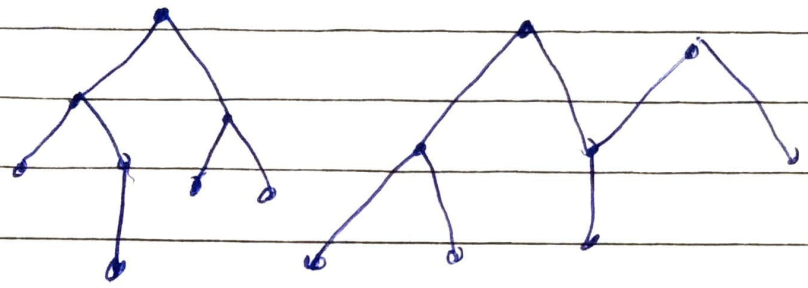
(1)



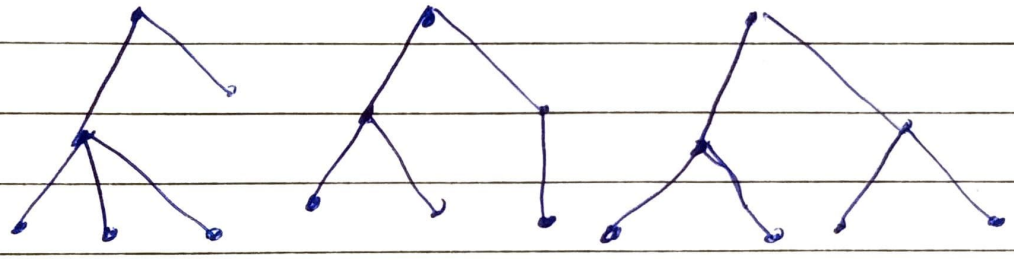
Find the levels of vertices A, H, F, M

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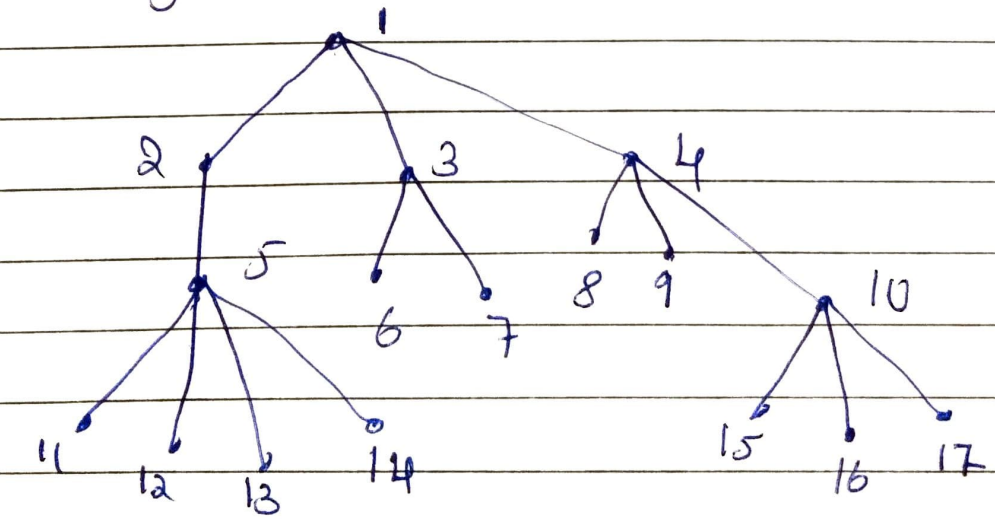
(2) Which of the following is a rooted tree?



(3) Which of the following is a binary tree? Complete binary tree?



(4) Write a Preorder & Postorder traversal for a tree.



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(5) Apply the merge sort to the list.

(i) −1, 7, 4, 11, 5, −8, 15, −3, −2, 6, 10, 3

(ii) 6, 2, 7, 3, 4, 9, 5, 1, 8.

(iii) −1, 0, 2, −2, 3, 6, −3, 5, 1, 4

(iv) −9, 6, 5, −3, 4, 2, −7, 6, −5, 10, −11, 0